

Advanced Financial Modeling: Generative AI Investment Analysis Project

Group Project

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Abstract

This project applies advanced financial modeling techniques to evaluate generative AI investment projects using capital budgeting methods. Students will select a publicly traded company, download historical financial data using Google Colab, and analyze a hypothetical generative AI investment project using four capital budgeting methodologies. The analysis includes comprehensive sensitivity analysis, real options valuation, Monte Carlo simulation, and visualization of results to support investment decision-making for AI initiatives.

1 Project Overview

1.1 Objective

The primary objective is to provide hands-on experience in financial modeling and capital budgeting analysis for generative AI investments using Python in Google Colab. Students will learn to:

- Download and process historical financial data
- Apply four capital budgeting methods to AI projects
- Conduct sensitivity analysis for technology investments
- Implement Real Options valuation for AI project flexibility
- Perform Monte Carlo simulation for AI project risk analysis

- Visualize and interpret results
- Make AI investment recommendations

1.2 Company Selection

Each student group will select one publicly traded company with:

- Sufficient historical financial data
- Clear business model and AI investment opportunities
- Relevance to generative AI applications (e.g., tech, healthcare, finance, retail)
- Examples: Microsoft, Google, Amazon, Nvidia, Salesforce, Adobe, etc.

2 Required Analysis Components

2.1 Data Collection and Project Setup

Students must download historical financial statements using Google Colab and provide:

1. Financial Data Collection:

- Income Statement data for the last 5 years
- Balance Sheet data for the last 5 years
- Cash Flow Statement data for the last 5 years
- Free Cash Flow calculation for investment sizing

2. Generative AI Project Definition:

- Project scope and objectives
- Expected AI applications (e.g., customer service, content generation, process automation)
- Implementation timeline (3-5 years)
- Key success metrics

3. Investment Parameters:

- Initial Investment = 1% of company's last year Free Cash Flow
- Annual cash inflow = initial investment $\times$ AI efficiency factor (company-specific)
- Project life = 5 years (typical for technology projects)
- Discount rate = company WACC + AI risk premium (2-4%)

2.2 Capital Budgeting Methods for AI Projects

Students must implement and analyze FOUR capital budgeting methods specifically for generative AI investments:

1. **The Payback Rule** - AI project risk assessment
2. **Net Present Value (NPV)** - Traditional value calculation
3. **The Profitability Index (PI)** - Resource allocation efficiency
4. **The Internal Rate of Return (IRR)** - Return comparison with hurdle rates

2.3 Methodology Explanation

For EACH capital budgeting method, students must provide:

2.3.1 Theoretical Foundation

- Mathematical formulation and equations
- Theoretical basis and assumptions
- Calculation methodology adapted for AI projects

2.3.2 Advantages and Limitations for AI Investments

- Key advantages of the method for technology projects
- Limitations in capturing AI-specific risks and opportunities
- Appropriate use cases for generative AI investments
- Comparison with other methods

2.4 Sensitivity Analysis for AI Projects

Students must conduct comprehensive sensitivity analysis on:

- AI adoption rates and user acceptance
- Technology cost reductions over time
- Competitive responses and market dynamics
- Regulatory changes affecting AI deployment
- Data quality and availability assumptions

2.5 Visualization Requirements

Create professional visualizations including:

- NPV profile charts with AI risk factors
- Sensitivity tornado diagrams for key AI variables
- Payback period comparisons across scenarios
- IRR distribution plots for different adoption rates
- AI implementation roadmap and cash flow timelines

3 Advanced Methods: Real Options and Monte Carlo Simulation for AI

3.1 Real Options Analysis for AI Investments

3.1.1 Assignment Objectives

- Understand the limitations of traditional NPV for AI projects
- Learn to identify and value real options in AI investments
- Implement binomial models for AI project flexibility
- Quantify the value of staged AI implementation

3.1.2 Required Tasks

Task 1: AI Real Options Identification

- a) Select your generative AI investment project
- b) Identify at least THREE types of real options embedded in the AI project:
 - **Option to Expand:** Scale AI to other business units
 - **Option to Abandon:** Terminate underperforming AI applications
 - **Option to Delay:** Wait for technology maturation
 - **Option to Switch:** Change AI models or vendors
 - **Option to Learn:** Pilot projects with option to scale
- c) For each option, provide:
 - Clear description of the AI-specific flexibility
 - Technical and business conditions for exercise
 - AI capability value drivers

Task 2: Binomial Model Implementation for AI Expansion

- a) Implement a multi-period binomial model in Python
- b) Value an AI expansion option with company-specific parameters:
 - Current AI project value: Based on your cash flow projections
 - Expansion investment: Additional AI infrastructure costs
 - Expansion factor: 1.5x-3x current value (AI scalability)
 - Time to decision: 1-2 years (typical AI evaluation period)
 - Volatility: 40-60% annually (high for emerging tech)
 - Risk-free rate: Current government bond rates
 - Number of periods: 4-8 (quarterly or semi-annual steps)
- c) Calculate and report:

- Risk-neutral probabilities for AI success
- Option value at each decision point
- Final expansion option value

Task 3: AI Abandonment Option Analysis

- a) Value an AI abandonment option with:
 - AI project value: Your calculated project value
 - Salvage value: Reusable components, data assets, talent
 - Time frame: 2-3 years (typical AI proof-of-concept period)
 - Volatility: 35-50% (technology project risk)
- b) Compare abandonment option value with and without the option
- c) Analyze the "technology risk insurance" value

Task 4: Strategic Implications for AI Investment

- a) Calculate Expanded NPV for AI project:

$$\text{Expanded NPV} = \text{Traditional NPV} + \text{AI Real Options Value}$$
- b) Discuss how real options analysis changes the AI investment decision
- c) Provide strategic recommendations for phased AI implementation

3.1.3 Deliverables for Real Options Assignment

- Python code implementing binomial models for AI options
- AI option valuation results and sensitivity analysis
- Strategic analysis of AI implementation flexibility
- Visualization of AI decision trees and option values

3.2 Monte Carlo Simulation for AI Project Risk

3.2.1 Assignment Objectives

- Understand probability distributions for AI project variables
- Learn to model uncertainty in AI investments
- Implement Monte Carlo simulation for AI risk assessment
- Analyze AI project risk-return trade-offs

3.2.2 Required Tasks

Task 1: AI Uncertainty Modeling

- a) Identify key uncertain variables in your AI investment project:
 - AI model accuracy and performance
 - User adoption rates and engagement
 - Data quality and availability
 - Technology cost evolution
 - Competitive AI advancements
- b) Define appropriate probability distributions for:
 - AI efficiency gains (Normal/Lognormal distribution)
 - Implementation timeline (Triangular distribution)
 - Adoption rates (Beta distribution)
 - Technology costs (Uniform distribution with trend)
- c) Justify your choice of distributions based on AI industry data

Task 2: Monte Carlo Implementation for AI NPV

- a) Implement a Monte Carlo simulation with 10,000 iterations
- b) Model AI project NPV using realistic uncertain inputs:
 - Initial AI investment: Based on 1% of FCF $\pm$ 20%

- Annual AI benefits: Normal(your projection, 30% std dev)
- AI adoption ramp: Triangular(20%, 50%, 80% year 1 adoption)
- Cost reduction: Normal(10%, 5%) annual efficiency gains
- Project life: Discrete{3:0.2, 5:0.5, 7:0.3} (technology refresh)
- Discount rate: Uniform(WACC-1%, WACC+3%) for AI risk

c) Calculate NPV distribution for AI project

Task 3: AI Project Risk Analysis

a) Calculate and report AI-specific risk metrics:

- Mean NPV and standard deviation
- Probability of negative NPV (AI project failure risk)
- Value at Risk (5th percentile) - worst-case scenarios
- 90% confidence interval for AI project value
- Probability of exceeding AI investment hurdles

b) Create probability distribution charts for AI project outcomes

Task 4: AI Scenario and Sensitivity Analysis

a) Perform tornado analysis to identify key AI value drivers

b) Test three AI-specific scenarios:

- Base case: Expected AI performance
- Optimistic case: Rapid adoption, high efficiency gains
- Pessimistic case: Slow adoption, technical challenges

c) Analyze correlation between AI success factors and project value

3.2.3 Deliverables for Monte Carlo Assignment

- Python code for AI project Monte Carlo simulation
- Statistical analysis of AI project risks
- Probability distribution charts for AI outcomes
- AI sensitivity analysis and key risk drivers
- Risk mitigation recommendations for AI implementation

3.3 Integrated Analysis: Combining Methods for AI Investment

3.3.1 Comprehensive AI Project Evaluation

- a) Apply all three methods to your generative AI investment:
 - Traditional capital budgeting (NPV, IRR, Payback, PI)
 - Real Options analysis for AI flexibility
 - Monte Carlo simulation for AI risk assessment
- b) Compare and contrast results from different methods
- c) Resolve conflicts between method recommendations
- d) Provide comprehensive AI investment recommendation

3.3.2 AI Investment Decision Framework

- a) Develop an AI-specific decision framework that incorporates:
 - Traditional financial metrics
 - Strategic flexibility value from real options
 - Technology risk assessment from Monte Carlo
 - AI capability development and learning value
- b) Create an AI investment scorecard with weighted criteria
- c) Provide phased AI implementation roadmap

4 Project Deliverables

4.1 Written Report

- Company background and AI investment rationale
- Generative AI project definition and scope
- Detailed methodology explanations adapted for AI

- Advantages and limitations of each method for technology investments
- Real Options analysis and AI flexibility valuation
- Monte Carlo simulation results and AI risk assessment
- Integrated AI investment analysis
- Strategic recommendations for AI implementation
- Conclusion and AI investment insights

5 AI Project Assumptions and Parameters

5.1 Generative AI Investment Specifics

- **Project Life:** 5 years (technology refresh cycle)
- **Initial Investment:** 1% of company's Free Cash Flow
- **AI Efficiency Factor:** Company-specific (0.25-0.40 typically)
- **Discount Rate:** Use one of these simplified approaches:
 - **Option 1:** Company's historical ROE (Return on Equity)
 - **Option 2:** Industry average cost of capital (10-15% for tech)
 - **Option 3:** Risk-free rate + 5-8% risk premium
 - **Option 4:** 12% as standard technology project rate
- **Implementation Timeline:** 6-18 months for initial deployment

5.2 Simplified Discount Rate Calculation

1. Find Company's ROE:

$$\text{ROE} = \frac{\text{Net Income}}{\text{Shareholders' Equity}}$$

Use the average ROE from last 3-5 years

2. Add AI Risk Premium:

$$\text{Discount Rate} = \text{ROE} + 3\%$$

3. Alternative Simple Method:

- Risk-free rate (10-year government bond): 4-5%
- Market risk premium: 5-6%
- AI technology premium: 3-4%
- **Total:** 12-15% discount rate

5.3 Recommended Simple Approach

- **Tech Companies:** 12-14% discount rate
- **Non-Tech Companies:** 10-12% discount rate
- **Conservative:** Use 15% for high uncertainty AI projects
- **Standard Baseline:** 12% for most generative AI investments